Lecture 17:

Introduction to VLSI On-Die Termination Circuits

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This Topic: Introduction to I/O Termination Circuits

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Introduction

- Proper termination of the I/O channel has been one of the most important design tasks in high-speed I/O circuits
- Channel terminations are critical to minimize the channel ringing, reflection and X-talk effects
- Closed loop On Die Termination (ODT) circuits are commonly used in VLSI high-speed I/O circuits to compensate for the PVT variations of the circuits
  - Also known as R-Comp circuit
- ODT are usually implemented on both Tx and Rx sides for better signal integrity (SI) performance
- One of key design task for high-speed I/O ODT designs is to minimize the pad capacitance at the Tx and Rx ports

Introduction

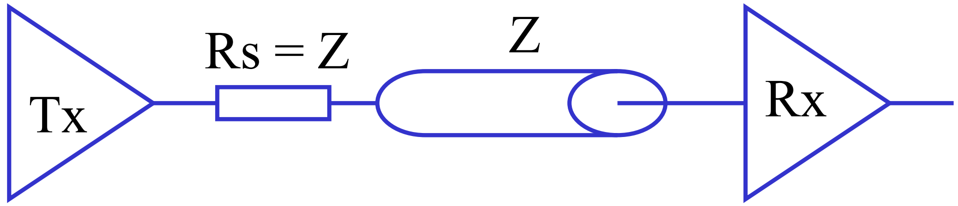
- Every I/O standard has its specified characteristic impedance
- The range of impedance varies depending on the chip family and bus architecture
  - Typical range are 25Ω to above 250Ω with 50Ω as commonly used value.
- Ringing happens in short improperly terminated transmission lines where the total length of all stubs on a line are not short enough or there is a mismatch in impedance along the transmission line such as a connector, this can cause false signal edge due to amplitude of ring.
- Reflections occur in long improperly terminated transmission lines and can cause timing problems due to the inherent property of voltage doubling on an open transmission line

ODT Circuit Architectures

- I/O Termination configurations
  - Parallel Termination
  - Serial Termination
  - Theveine Termination
  - Active Load Termination
  - etc.
- Terminating Locations
  - Receiver or Far end termination
  - Source or Near end termination
  - Star or Center Termination
  - etc.

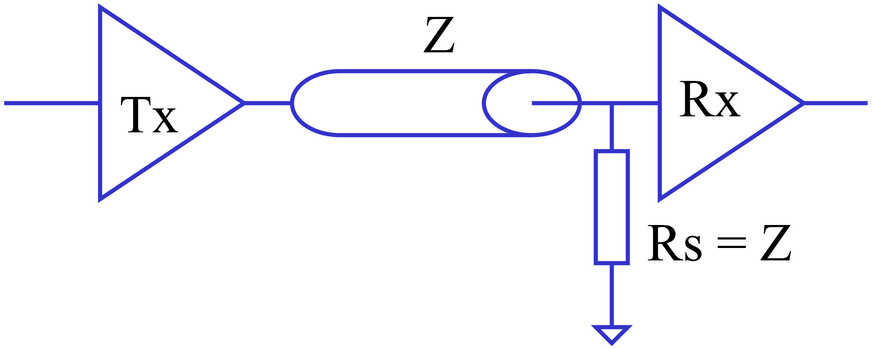
Serial Termination

- Termination resistor is located at the output of the driver
- This termination configuration is also known as near end termination



Parallel Termination

- Termination resistor is located at the input of the receiver
- This termination configuration is also known as far end termination



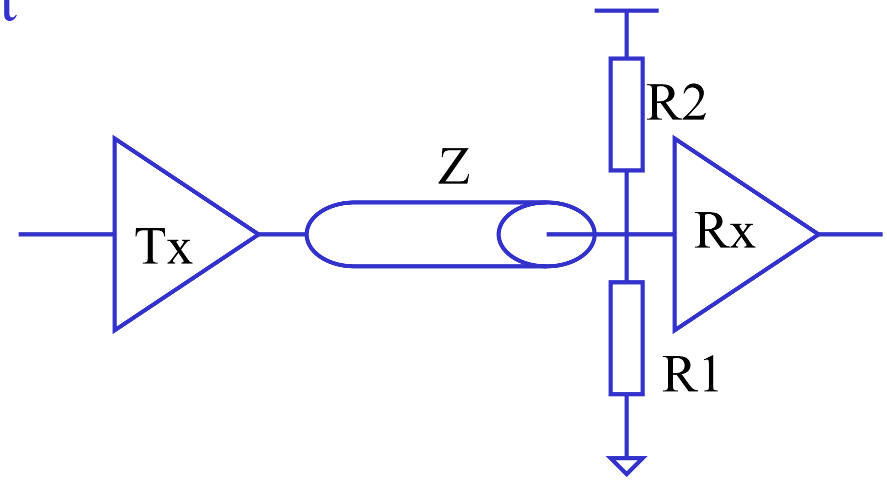
Thevenin Termination

- Termination resistors are located at the input node of Rx
  - Voltage operation point

$$V_{Cm} = VccR_2/(R_1+R_2)$$

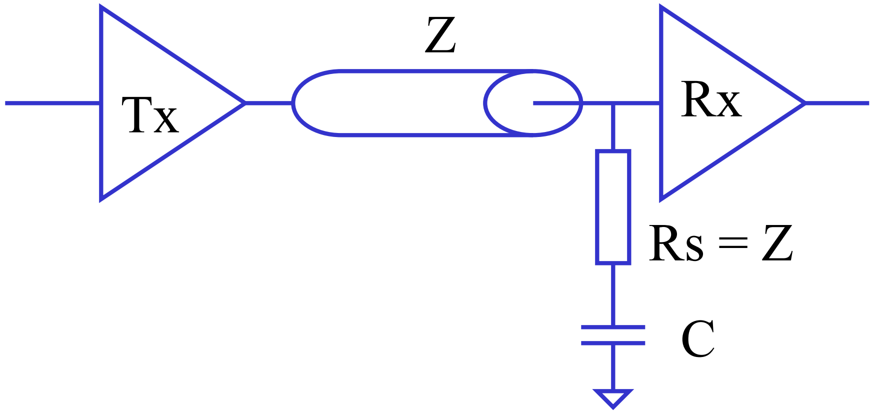
  - Termination resistor

$$Z = R_1R_2/(R_1+R_2)$$

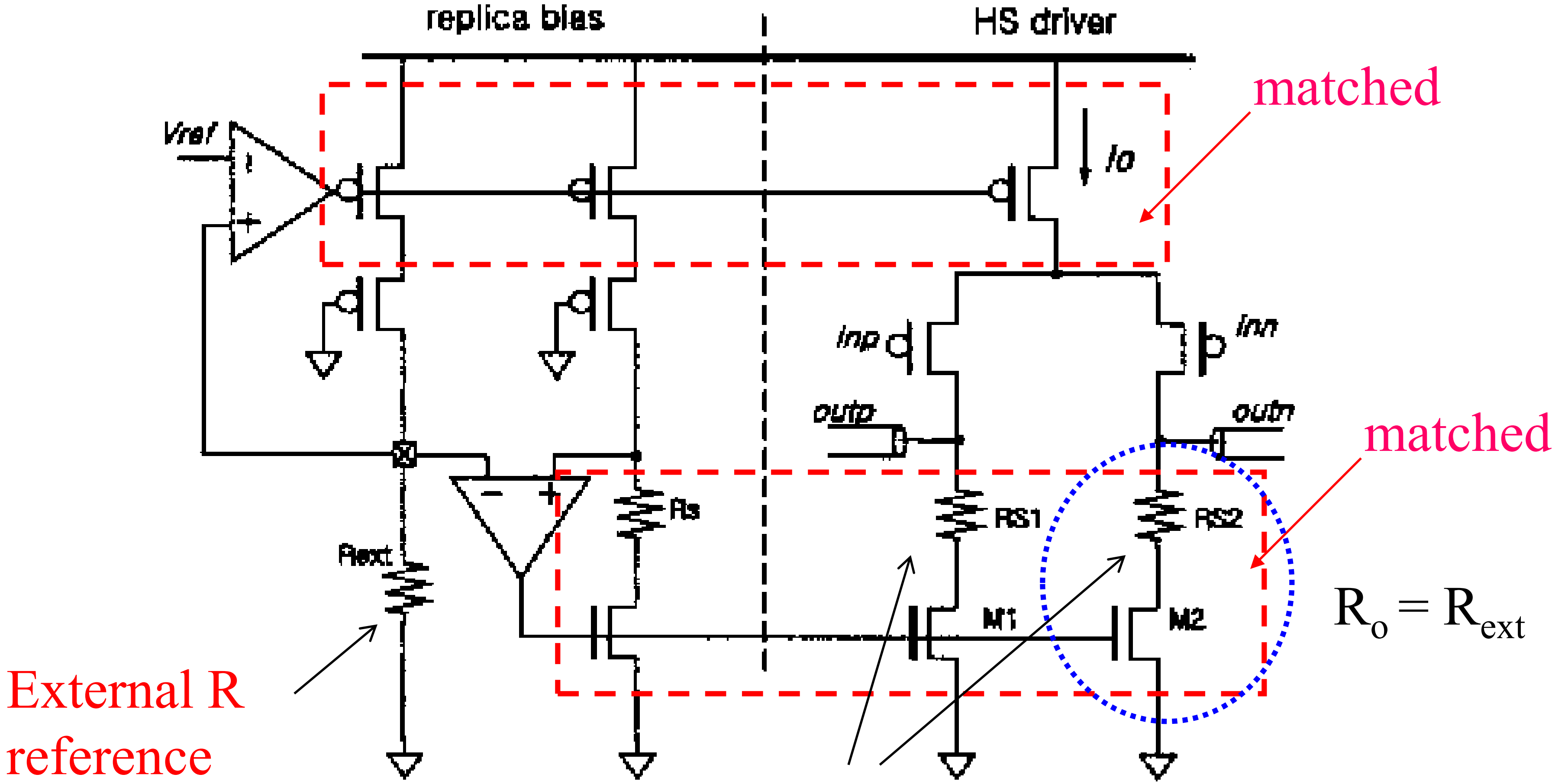


AC Termination

- A Ac termination adds a Capacitor in series with the termination resistor
  - AC short (terminated)
  - DC open (unterminated)



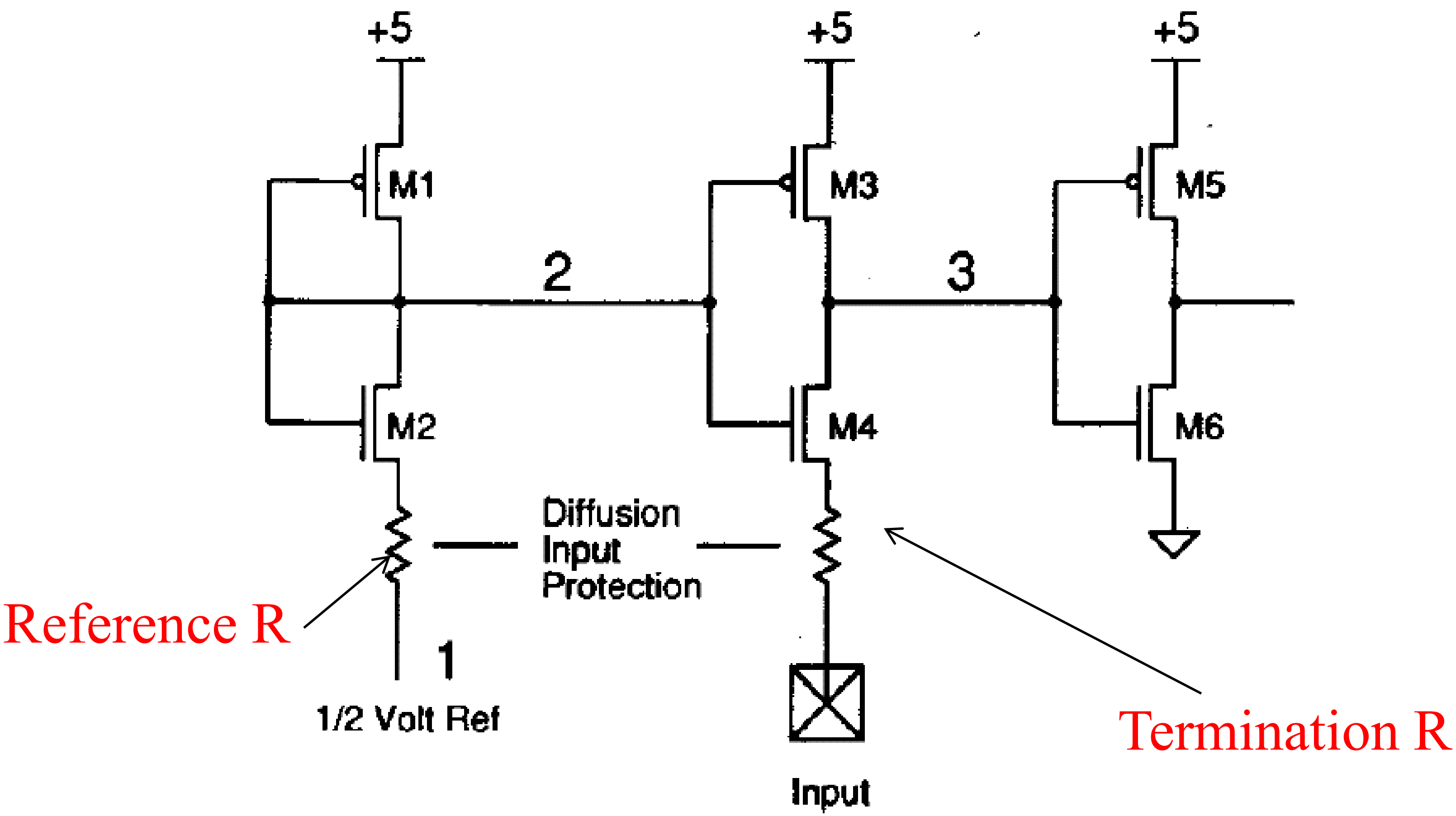
VLSI ODT Circuit Implementations



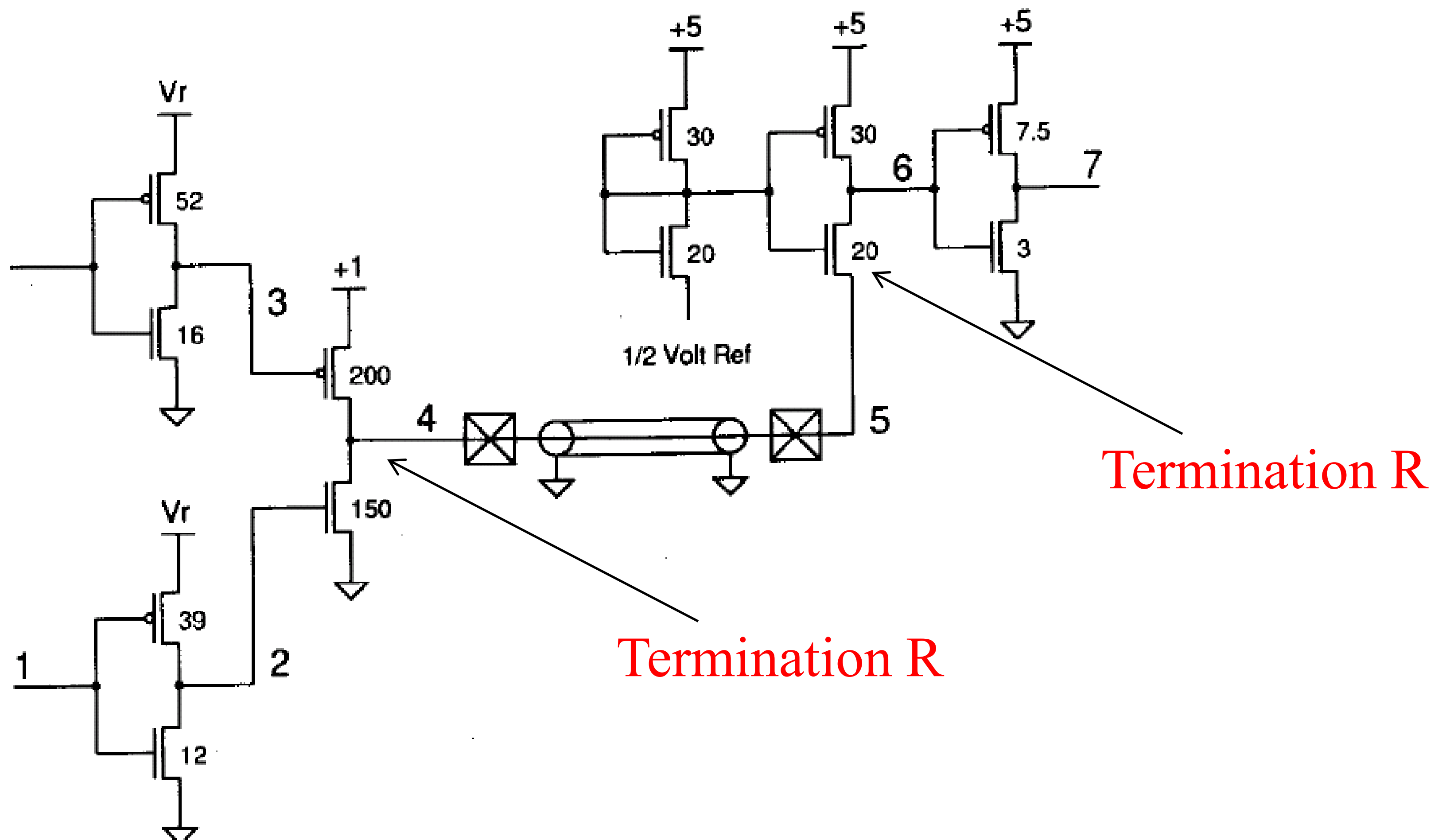
$R_o = R_{ext}$

VLSI ODT Circuit Implementations

- Single side driver



VLSI ODT Circuit Implementations



VLSI ODT Circuit Implementations

- Programmable Tx output impedance

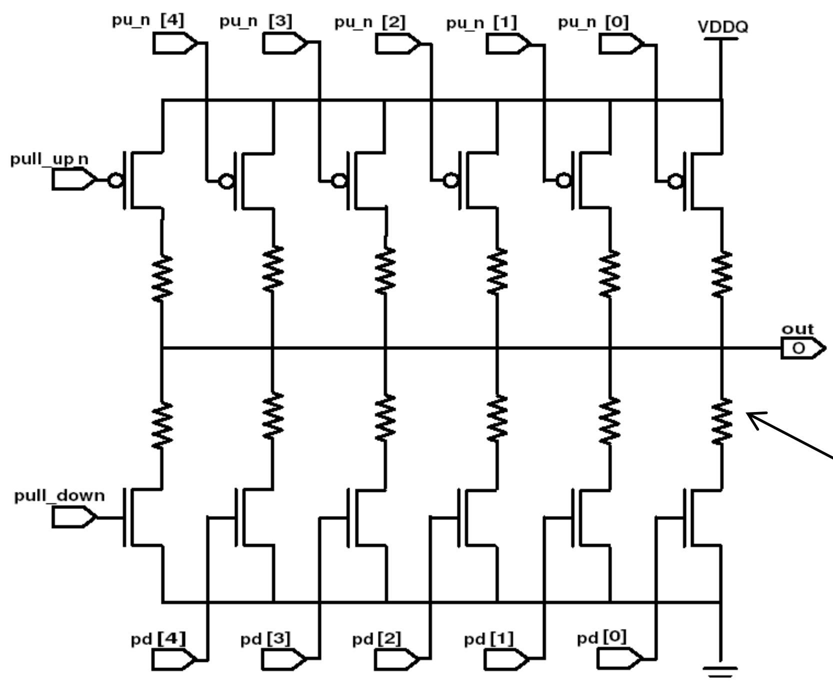


Fig. 5. Poly resistance I/O driver

Termination R

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VLSI ODT Circuit Implementations

- Programmable Tx output impedance

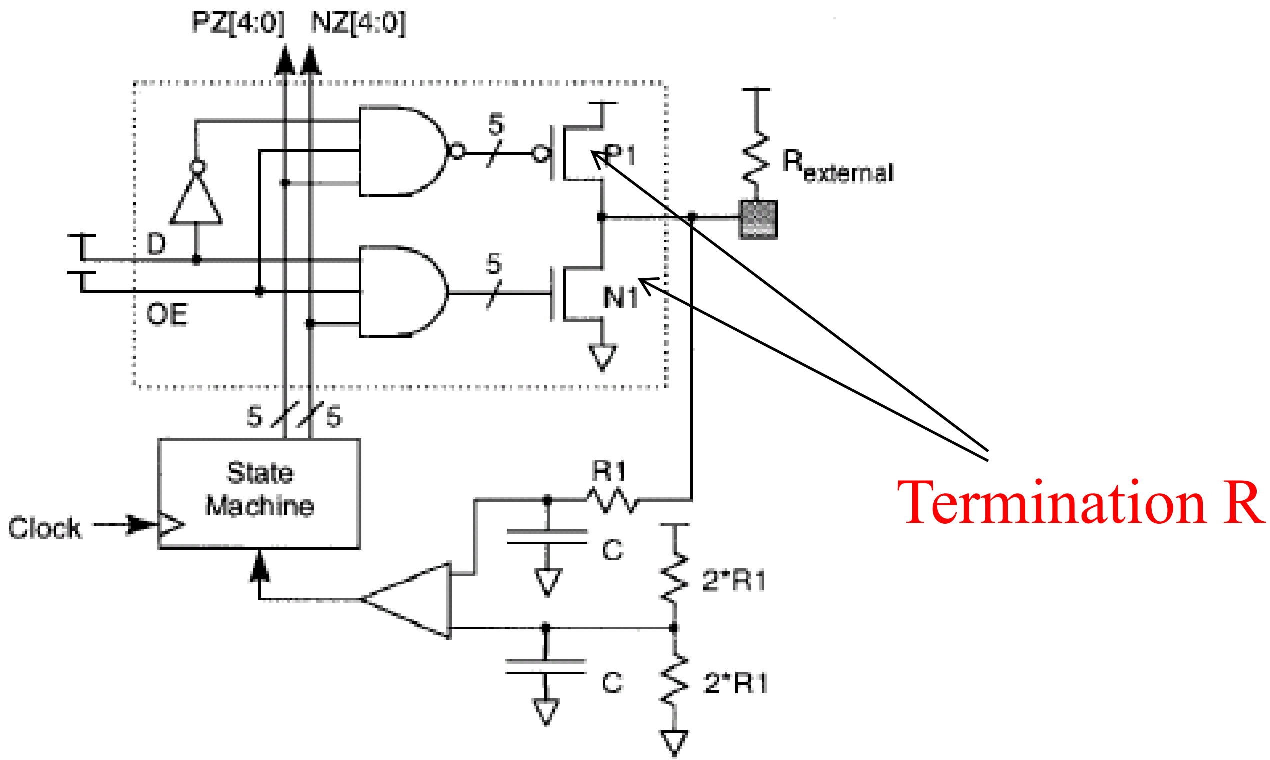
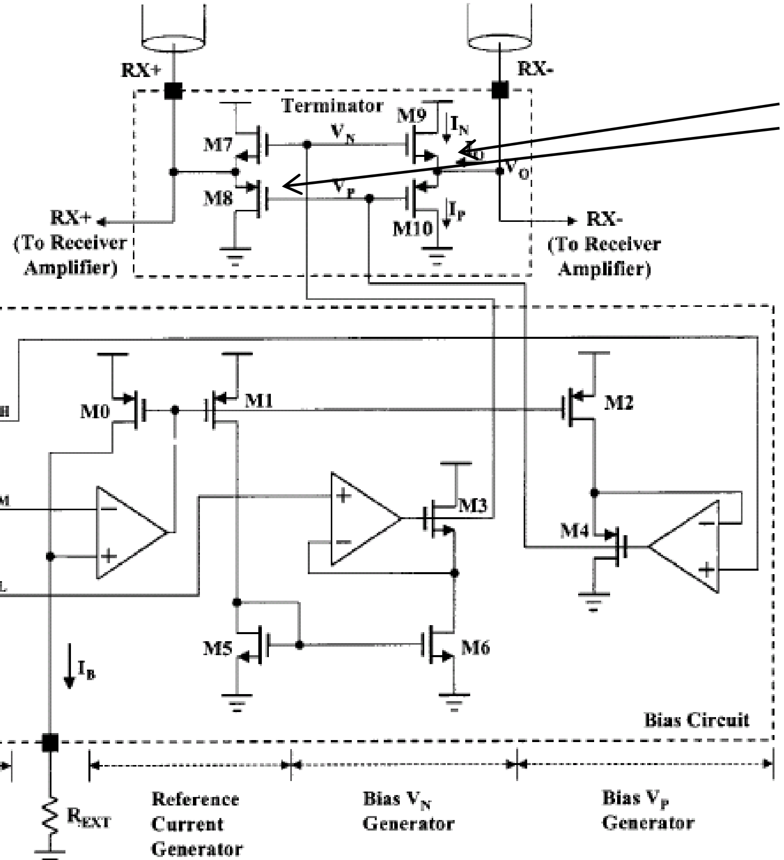


Fig. 3. Impedance control loop

VLSI ODT Circuit Implementations

- Programmable Tx output impedance

Termination R



VLSI ODT Circuit Implementations

- Differential Termination